



How similar are two strings?

- Spell correction

- The user typed “graffe”

Which is closest?

- graf
 - graft
 - grail
 - giraffe

- Computational Biology

- Align two sequences of nucleotides

```
AGGCTATCACCTGACCTCCAGGCCGATGCCC
TAGCTATCACGACCGCGGTCGATTGCCCCGAC
```

- Resulting alignment:

```
-AGGCTATCACCTGACCTCCAGGCCGA--TGCCC---
TAG-CTATCAC--GACCGC--GGTCGATTGCCCCGAC
```

- Also for Machine Translation, Information Extraction, Speech Recognition



Edit Distance

- The minimum edit distance between two strings
- Is the minimum number of editing operations
 - Insertion
 - Deletion
 - Substitution
- Needed to transform one into the other



Minimum Edit Distance

- Two strings and their **alignment**:

I	N	T	E	*	N	T	I	O	N
*	E	X	E	C	U	T	I	O	N



Minimum Edit Distance

I N T E * N T I O N

| | | | | | | | | |

* E X E C U T I O N

d s s i s

- If each operation has cost of 1
 - Distance between these is 5
- If substitutions cost 2 (Levenshtein)
 - Distance between them is 8



Alignment in Computational Biology

- Given a sequence of bases

AGGCTATCACCTGACCTCCAGGCCGATGCCC
TAGCTATCACGACCGCGGGTCGATTGCCCCGAC

- An alignment:

–AGGCTATCACCTGACCTCCAGGCCGA–TGCCC–
TAG–CTATCAC–GACCGC–GGTCGATTGCCCCGAC

- Given two sequences, align each letter to a letter or gap



Other uses of Edit Distance in NLP

- Evaluating Machine Translation and speech recognition

R Spokesman confirms senior government adviser was shot

H Spokesman said the senior adviser was shot dead

S

I

D

I

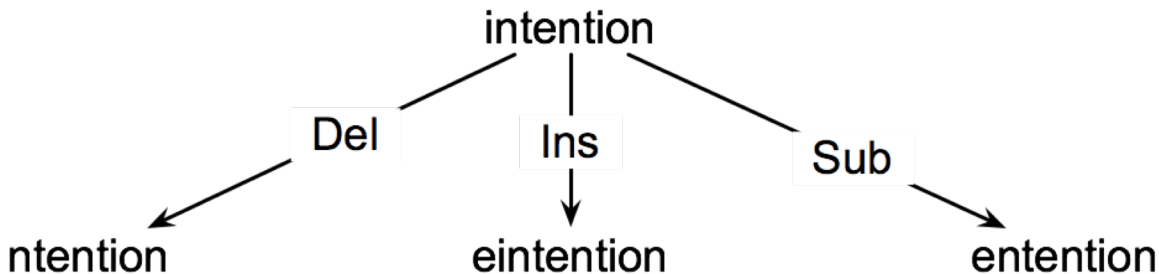
- Named Entity Extraction and Entity Coreference

- IBM Inc. announced today
- IBM profits
- Stanford President John Hennessy announced yesterday
- for Stanford University President John Hennessy



How to find the Min Edit Distance?

- Searching for a path (sequence of edits) from the start string to the final string:
 - **Initial state:** the word we're transforming
 - **Operators:** insert, delete, substitute
 - **Goal state:** the word we're trying to get to
 - **Path cost:** what we want to minimize: the number of edits





Minimum Edit as Search

- But the space of all edit sequences is huge!
 - We can't afford to navigate naïvely
 - Lots of distinct paths wind up at the same state.
 - We don't have to keep track of all of them
 - Just the shortest path to each of those revisited states.



Defining Min Edit Distance

- For two strings
 - X of length n
 - Y of length m
- We define $D(i,j)$
 - the edit distance between $X[1..i]$ and $Y[1..j]$
 - i.e., the first i characters of X and the first j characters of Y
 - The edit distance between X and Y is thus $D(n,m)$

Minimum Edit Distance

Computing Minimum Edit Distance



Dynamic Programming for Minimum Edit Distance

- **Dynamic programming:** A tabular computation of $D(n,m)$
- Solving problems by combining solutions to subproblems.
- Bottom-up
 - We compute $D(i,j)$ for small i,j
 - And compute larger $D(i,j)$ based on previously computed smaller values
 - i.e., compute $D(i,j)$ for all i ($0 < i < n$) and j ($0 < j < m$)



Defining Min Edit Distance (Levenshtein)

- Initialization

$$D(i, 0) = i$$

$$D(0, j) = j$$

- Recurrence Relation:

For each $i = 1 \dots M$

For each $j = 1 \dots N$

$$D(i, j) = \min \begin{cases} D(i-1, j) + 1 & \text{insert str2[j] at i} \\ D(i, j-1) + 1 & \text{insert str[i] at j} \\ D(i-1, j-1) + \begin{cases} 2; & \text{if } X(i) \neq Y(j) \\ 0; & \text{if } X(i) = Y(j) \end{cases} \end{cases}$$

substitution cost is 2

- Termination:

$D(N, M)$ is distance



The Edit Distance Table

N	9									
O	8									
I	7									
T	6									
N	5									
E	4									
T	3									
N	2									
I	1									
#	0	1	2	3	4	5	6	7	8	9
	#	E	X	E	C	U	T	I	O	N



The Edit Distance Table

N	9									
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I	1									
#	0	1	2	3	4	5	6	7	8	9
	#	E	X	E	C	U	T	I	O	N

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } S_1(i) \neq S_2(j) \\ 0; & \text{if } S_1(i) = S_2(j) \end{cases} \end{cases}$$

 A red arrow points from the top right towards the cell containing 'I' in the first column and '1' in the second column of the table.



Edit Distance

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } S_1(i) \neq S_2(j) \\ 0; & \text{if } S_1(i) = S_2(j) \end{cases} \end{cases}$$

N	9									
O	8									
I	7									
T	6									
N	5									
E	4									
T	3									
N	2									
I	1									
#	0	1	2	3	4	5	6	7	8	9
	#	E	X	E	C	U	T	I	O	N



The Edit Distance Table

N	9	8	9	10	11	12	11	10	9	8
O	8	7	8	9	10	11	10	9	8	9
I	7	6	7	8	9	10	9	8	9	10
T	6	5	6	7	8	9	8	9	10	11
N	5	4	5	6	7	8	9	10	11	10
E	4	3	4	5	6	7	8	9	10	9
T	3	4	5	6	7	8	7	8	9	8
N	2	3	4	5	6	7	8	7	8	7
I	1	2	3	4	5	6	7	6	7	8
#	0	1	2	3	4	5	6	7	8	9
	#	E	X	E	C	U	T	I	O	N



Minimum Edit Distance

Backtrace for
Computing Alignments



Computing alignments

- Edit distance isn't sufficient
 - We often need to **align** each character of the two strings to each other
- We do this by keeping a “backtrace”
- Every time we enter a cell, remember where we came from
- When we reach the end,
 - Trace back the path from the upper right corner to read off the alignment



Edit Distance

$$D(i,j) = \min \begin{cases} D(i-1,j) + 1 \\ D(i,j-1) + 1 \\ D(i-1,j-1) + \begin{cases} 2; & \text{if } S_1(i) \neq S_2(j) \\ 0; & \text{if } S_1(i) = S_2(j) \end{cases} \end{cases}$$

N	9									
O	8									
I	7									
T	6									
N	5									
E	4									
T	3									
N	2									
I	1									
#	0	1	2	3	4	5	6	7	8	9
	#	E	X	E	C	U	T	I	O	N



MinEdit with Backtrace

n	9	↓ 8	↙←↓ 9	↙←↓ 10	↙←↓ 11	↙←↓ 12	↓ 11	↓ 10	↓ 9	↙ 8	
o	8	↓ 7	↙←↓ 8	↙←↓ 9	↙←↓ 10	↙←↓ 11	↓ 10	↓ 9	↙ 8	← 9	
i	7	↓ 6	↙←↓ 7	↙←↓ 8	↙←↓ 9	↙←↓ 10	↓ 9	↙ 8	← 9	← 10	
t	6	↓ 5	↙←↓ 6	↙←↓ 7	↙←↓ 8	↙←↓ 9	↙ 8	← 9	← 10	←↓ 11	
n	5	↓ 4	↙←↓ 5	↙←↓ 6	↙←↓ 7	↙←↓ 8	↙←↓ 9	↙←↓ 10	↙←↓ 11	↙↓ 10	
e	4	↙ 3	← 4	↙← 5	← 6	← 7	←↓ 8	↙←↓ 9	↙←↓ 10	↓ 9	
t	3	↙←↓ 4	↙←↓ 5	↙←↓ 6	↙←↓ 7	↙←↓ 8	↙ 7	←↓ 8	↙←↓ 9	↓ 8	
n	2	↙←↓ 3	↙←↓ 4	↙←↓ 5	↙←↓ 6	↙←↓ 7	↙←↓ 8	↓ 7	↙←↓ 8	↙ 7	
i	1	↙←↓ 2	↙←↓ 3	↙←↓ 4	↙←↓ 5	↙←↓ 6	↙←↓ 7	↙ 6	← 7	← 8	
#	0	1	2	3	4	5	6	7	8	9	
	#	e	x	e	c	u	t	i	o	n	



Adding Backtrace to Minimum Edit Distance

- Base conditions:

$$D(i, 0) = i$$

$$D(0, j) = j$$

Termination:

$D(N, M)$ is distance

- Recurrence Relation:

For each $i = 1 \dots M$

For each $j = 1 \dots N$

$$D(i, j) = \min \begin{cases} D(i-1, j) + 1 \\ D(i, j-1) + 1 \\ D(i-1, j-1) + \end{cases}$$

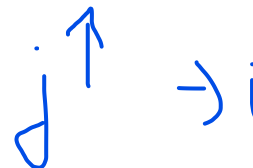
$$\begin{cases} 2; & \text{if } X(i) \neq Y(j) \\ 0; & \text{if } X(i) = Y(j) \end{cases}$$

$$\text{ptr}(i, j) = \begin{cases} \text{LEFT} & \text{insertion} \\ \text{DOWN} & \text{deletion} \\ \text{DIAG} & \text{substitution} \end{cases}$$

deletion

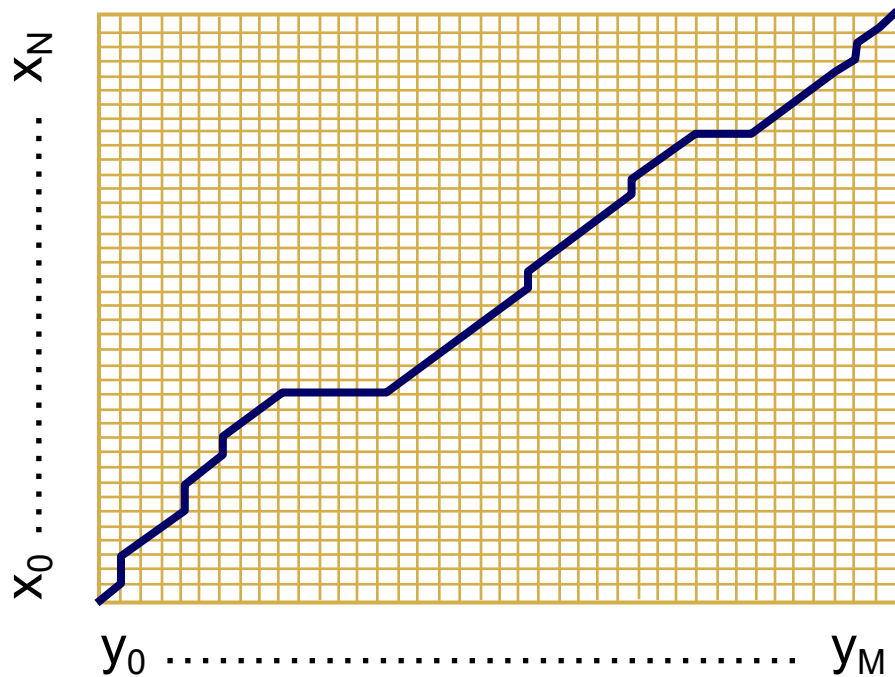
insertion

substitution





The Distance Matrix



Every non-decreasing path

from $(0,0)$ to (M, N)

corresponds to
an alignment
of the two sequences

An optimal alignment is composed
of optimal subalignments



Result of Backtrace

- Two strings and their **alignment**:

I	N	T	E	*	N	T	I	O	N
*	E	X	E	C	U	T	I	O	N



Performance

- Time:

$O(nm)$

- Space:

$O(nm)$

- Backtrace

$O(n+m)$



Weighted Edit Distance

- Why would we add weights to the computation?
 - Spell Correction: some letters are more likely to be mistyped than others
 - Biology: certain kinds of deletions or insertions are more likely than others





Weighted Min Edit Distance

- Initialization:

$$D(0, 0) = 0$$

$$D(i, 0) = D(i-1, 0) + \text{del}[x(i)]; \quad 1 < i \leq N$$

$$D(0, j) = D(0, j-1) + \text{ins}[y(j)]; \quad 1 < j \leq M$$

- Recurrence Relation:

$$D(i, j) = \min \begin{cases} D(i-1, j) + \text{del}[x(i)] \\ D(i, j-1) + \text{ins}[y(j)] \\ D(i-1, j-1) + \text{sub}[x(i), y(j)] \end{cases}$$

- Termination:

$D(N, M)$ is distance

